Bestseller Prediction Analysis: Final Report

Name: Dimakatso Galalelo

Surname: Tladi

Student number: ST10345238

Module code: PDAN8412 Part 2

Due Date: 21 October 2026

1. Executive Summary

Github Link

The objective of this project was to build a predictive machine learning model from scratch to determine whether a book has the potential to become a bestseller. Using the Amazon Top 500 Bestsellers as our "success" examples and a sample of general books from Google Books as our "unsuccessful" examples, we built a custom prediction tool.

The model demonstrated strong performance, correctly predicting the outcome for 88.3% of books it had never seen before. By analyzing the model's findings, we uncovered clear, actionable patterns: books with shorter titles, fiction categories, and multiple genre tags are significantly more likely to achieve bestseller status. This report outlines the data preparation, the patterns we discovered, the model's performance, and strategic recommendations for the publishing house.

2. Data Cleaning & Preparation

2.1 Protecting the Model from "Cheating" (Data Leakage)

In predictive modeling, there is a risk of the model "cheating" by learning from data that directly reveals the answer. Academic researchers Kaufman, Rosset and Perlich (2011) refer to this as "data leakage," which leads to false confidence in the model. To prevent this:

- **Removing Sales Metrics:** We removed variables like Amazon BSR (Best Seller Rank), Reviews, and Weeks on List. If we kept these, the model would simply read the sales rank to know if a book was a bestseller, rather than learning what actually makes a bestseller.
- **Filtering Keywords:** Any book in our "unsuccessful" group that had the word "bestseller" or "award" in its title or description was removed, ensuring the model didn't just look for a lucky keyword.
- **Removing Identifiers:** ISBNs and web links were removed so the model couldn't memorize specific books.

2.2 Handling Missing Data

The Google Books dataset had some pretty big problems, like the price info was missing for 80% of the books. According to research done by Alruhaymi and Kim (2021), when you try to guess or fill in missing data, it can mess things up and create a lot of bias. Since these missing pieces were connected to where the data came from (Google) and not the actual books themselves, we decided it was better to stick with only the complete and trustworthy information instead of making guesses.

2.3 Creating Comparable Features

Because the Amazon and Google datasets were formatted completely differently, we had to create a common language. We engineered new features based on the book's text:

- Title Length: Number of words in the title.
- Author Name Length: Number of words in the author's name.
- Publisher Name Length: Number of words in the publisher's name.
- Genre: Whether the book was Fiction or Non-Fiction, and how many genres it was tagged with.

2.4 Creating a Balanced Dataset

The final dataset combined 500 Amazon bestsellers with 1,000 Google Books, creating a 2:1 ratio. Because this is an imbalanced dataset (fewer bestsellers than non-bestsellers), we used a technique called "stratified sampling" to ensure both our training and testing data contained the same 2:1 ratio (He and Garcia, 2009).

3. Exploratory Data Analysis (EDA)

3.1 Feature Distributions

Our visual analysis showed that most books have relatively short titles (under 5 words), with a few outliers having exceptionally long titles.

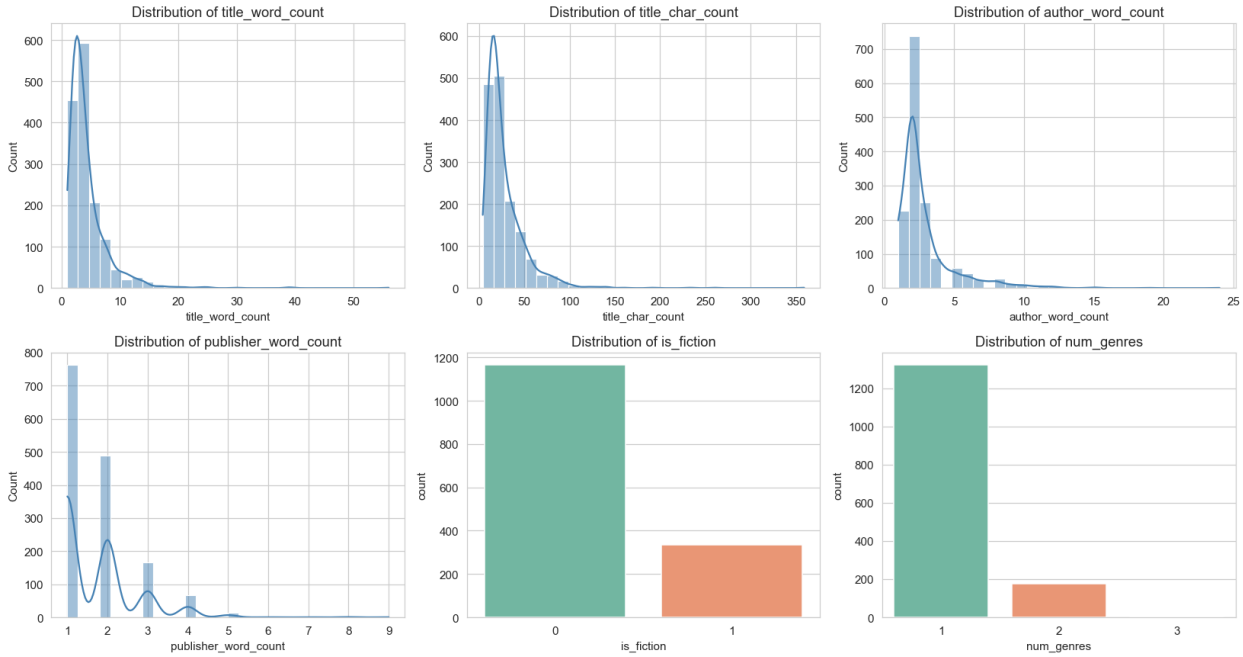


Figure 1: Distribution of Book Features. The charts show the frequency of title lengths, author name lengths, and the split between fiction and non-fiction.

3.2 What Makes a Bestseller?

We compared the successful books (bestsellers) against the unsuccessful ones to see how they differed.

- Fiction is King: Bestsellers have a significantly higher proportion of fiction books. This matches industry reports showing that adult fiction is currently the biggest driver of book sales (NielsenIQ BookData and GfK Entertainment, 2026).
- Shorter Titles Win: Bestsellers tend to have shorter, punchier titles. Marketing research consistently shows that shorter headlines and titles drive higher engagement (PR Newswire, 2025).
- Multiple Genres Help: Bestsellers are often tagged with more than one genre, suggesting that broader categorization helps discoverability.

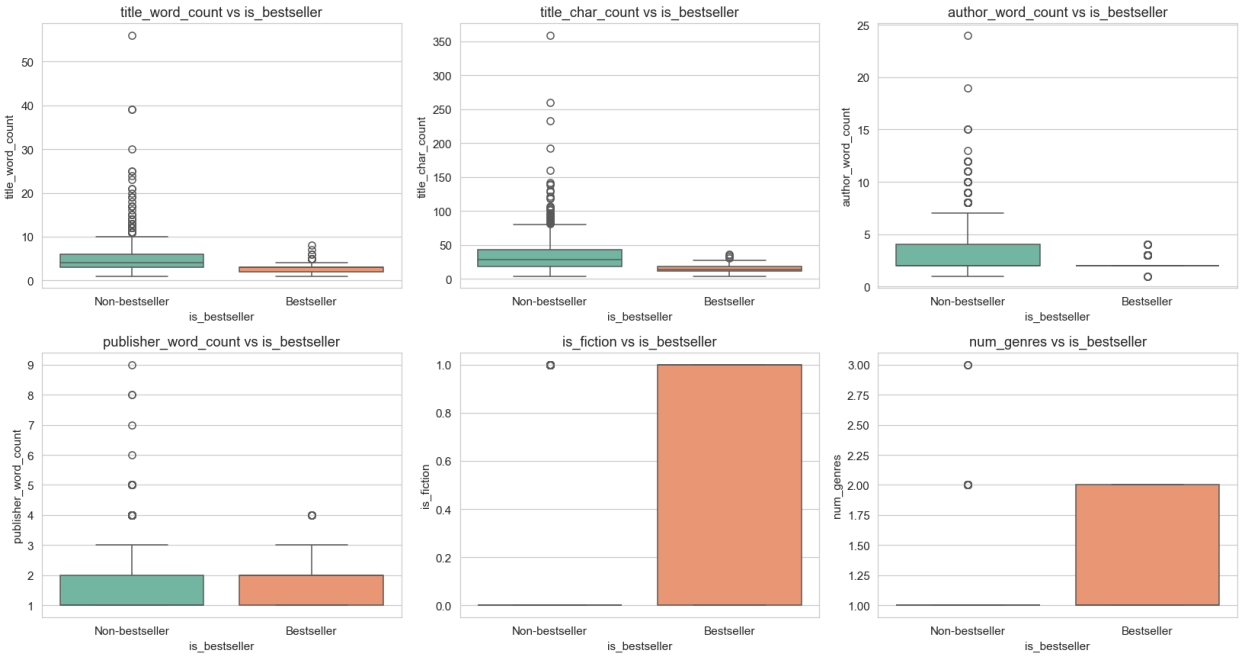


Figure 2: Comparing Features of Bestsellers vs. Non-Bestsellers. The charts show that bestsellers typically have shorter titles and a higher likelihood of being fiction.

3.3 Removing Redundant Information

We found that "Title Character Count" and "Title Word Count" were measuring the exact same thing. To keep the model simple and efficient, we removed the character count and kept the word count.

4. Model Training

4.1 How the Model Works

We built a custom Logistic Regression model from scratch. In simple terms, this model looks at a book's features (like title length and genre) and assigns a "weight" to each one. It then adds up all the weights to output a probability score (from 0% to 100%) indicating how likely the book is to be a bestseller (Hosmer & Lemeshow, 2000; Bishop, 2006).

4.2 Building from Scratch

We used Python's NumPy library Hosmer and Lemeshow (2000) to write the mathematical formulas that allow the model to learn. The model learns by repeatedly looking at the data, making predictions, calculating its error, and adjusting its internal weights to reduce that error (Goodfellow, Bengio and Courville, 2016). We also built in

the ability to apply "regularization" (L1 and L2), which prevents the model from overcomplicating its rules.

4.3 Training Setup

We split our data into an 80% training set (1,200 books) and a 20% testing set (300 books). We scaled the data so that large numbers (like word counts) didn't overwhelm smaller numbers (like binary yes/no genres), a standard best practice in data science. Finally, we set a "random seed" so that our results can be perfectly reproduced by anyone reviewing our work.

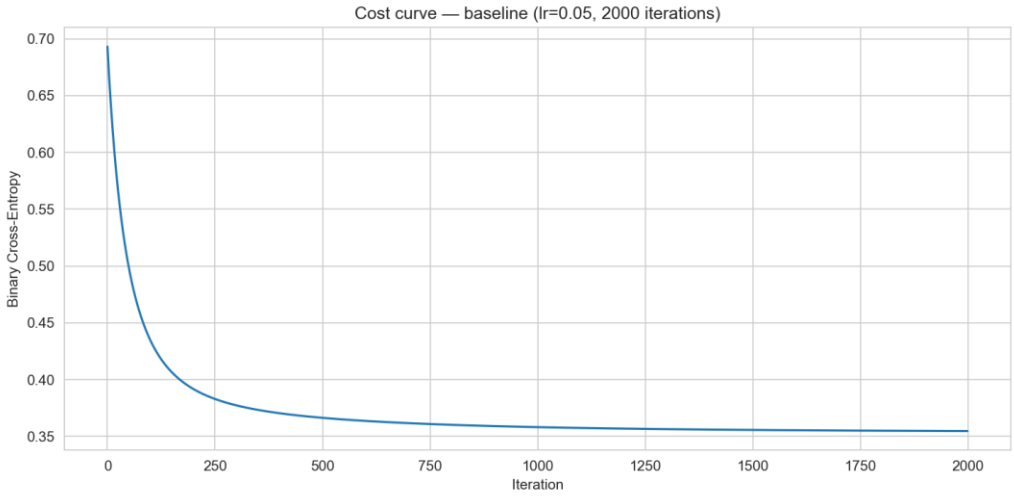


Figure 3: Learning Curve. The smooth downward line shows the model successfully learning and reducing its error over 2,000 iterations without getting confused.

5. Model Evaluation & Interpretation

5.1 How We Measured Success

Because we had more non-bestsellers than bestsellers, overall accuracy alone could be misleading. Instead, we used metrics recommended for business analysis (Sokolova and Lapalme, 2009):

- **Accuracy:** The overall percentage of correct predictions.
- **Precision and Recall:** How often the model was right when it predicted a bestseller, and how many actual bestsellers it managed to find.
- **Predictive Power (ROC-AUC):** This measures how well the model can separate the "successes" from the "failures" on a scale of 0.5 (random guess) to 1.0

(perfect prediction). A tool called the ROC Curve visually represents this (Fawcett, 2006).

5.2 Baseline Performance

The model achieved an outstanding 88.3% accuracy and a Predictive Power (ROC-AUC) of 0.928. This means the model has a very strong ability to rank a random bestseller higher than a random non-bestseller.

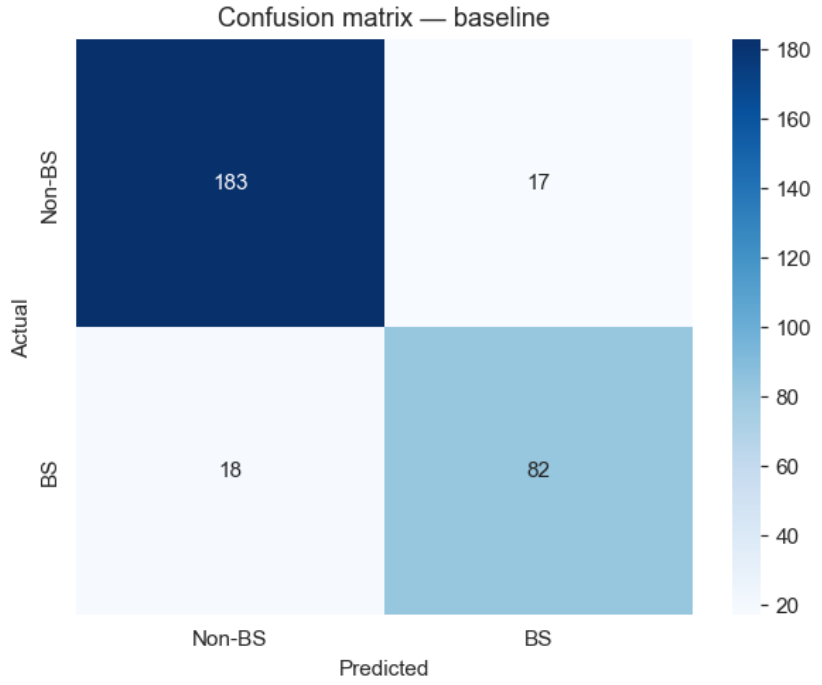


Figure 4: Confusion Matrix. This shows exactly how many books were correctly and incorrectly classified by the model.

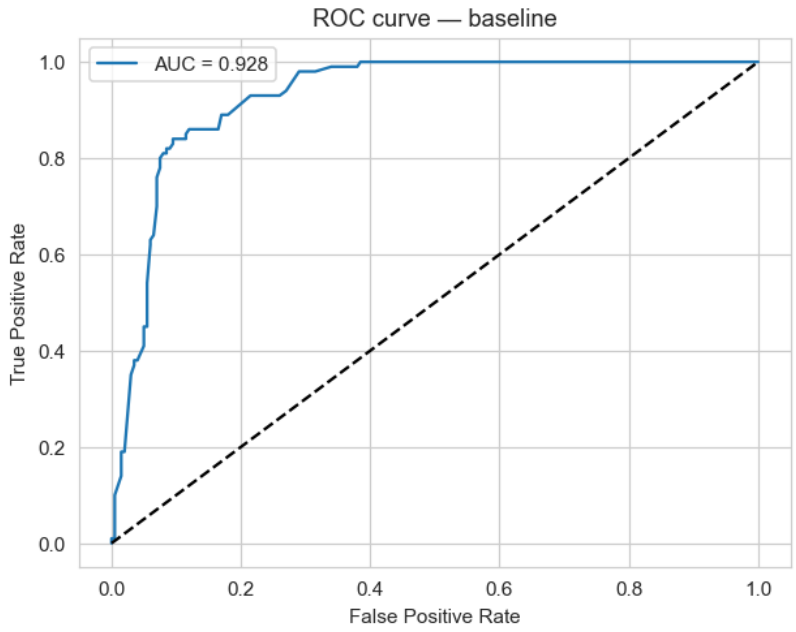


Figure 5: ROC Curve. The area under the curve (0.928) confirms the model's high predictive power.

5.3 Stress-Testing the Model

We retrained the model using different learning rates, different amounts of training time, and different mathematical constraints. The accuracy remained incredibly stable (between 88.0% and 88.3%). This is excellent news for the business, as it shows the model's predictions are robust and not easily broken (Geman, Bienenstock, and Doursat,1992).

5.4 Key Drivers of Bestseller Success

Feature	Impact	Business Interpretation
Title Word Count	Strongly Negative	Long titles decrease the chance of success. Concise titles are crucial.
Is Fiction	Strongly Positive	Fiction books have a massive advantage over non-fiction.
Number of Genres	Positive	Books tagged with multiple genres are more discoverable and more likely to succeed.
Author Word Count	Negative	Longer author names (or multiple authors) slightly reduce the chance of success.

6. Limitations & Caveats

6.1 The "Apples vs. Oranges" Problem (Domain Shift)

The most important limitation is that our "successes" came from Amazon (a retail platform curated by sales activity, updated daily), while our "failures" came from Google Books (a general database). In data science, this is called "Dataset Shift" (Quiñonera-Candela, et al., 2009). This means the model may have learned some differences in how Amazon and Google format their data, rather than purely what makes a bestseller. Therefore, the 88% accuracy is likely an upper bound and should be treated as a very strong screening tool, not an absolute guarantee.

6.2 Missing the "Real World" Ratio

In the real world, far fewer than 1 in 3 books become bestsellers. Our dataset artificially balanced the classes to help the model learn. This is a standard modeling technique, but it means the model's confidence scores should be treated as relative rankings rather than absolute probabilities.

6.3 Missing Features

The model only looks at text metadata. It cannot see the book cover, the author's reputation, or the marketing budget all of which heavily influence bestseller status. Future versions should incorporate author history and cover imagery.

7. Strategic Recommendations for the Client

1. Enforce Title Length Guidelines: The data is clear; shorter titles perform better. The editorial and marketing teams should treat title length as a critical design constraint.
2. Maximize Genre Tagging: Ensure every book is tagged with all relevant genres. Multi-genre tagging significantly improves the model's prediction of success.
3. Prioritize Fiction: Given fiction's strong positive weight, the publishing house should ensure fiction receives the majority of marketing and acquisition focus.
4. Use this as a Screening Tool: Because of the "Apples vs. Oranges" limitation, this model should be used to prioritize manuscripts for human review, not to make final acquisition decisions automatically.
5. Invest in Better Data: The single most valuable next step would be to gather historical data on books that did not become bestsellers from the same platform (Amazon books that never made the list). This would eliminate the data bias and create an even more accurate model.

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